Physical Chemistry Cume

1) The mechanism for the halogenation of a hydrocarbon (RH) using bromine (Br_2) is proposed to follow the following mechanism

$$Br_2 \xrightarrow{k_1} 2Br\bullet$$

$$Br\bullet + RH \xrightarrow{k_2} R\bullet + HBr$$

$$R\bullet + Br_2 \xrightarrow{k_3} RBr + Br\bullet$$

$$Br\bullet + R\bullet \xrightarrow{k_4} RBr$$

(a) Write down the overall balanced equation for this reaction.
(20 points)

(b) Determine the rate law predicted by this mechanism, stating any assumptions you make.
(50 points)

(c) Explain how you would go about proving or disproving this mechanism.
(30 points)